Part A – Learning Objective 3.2: I can derive the array factor for an arbitrary linear array and apply pattern multiplication.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Reading a sweep.** A lab group sweeps the PHASER with all eight elements active and a source at boresight. Working at 10.3 GHz with element spacing $d = 14$ mm, they record the trace below. The sweep step is $2.8125°$ and the peak reads -4.1 dBFS.

θ (°)	0.0	2.8	5.6	8.4	11.3	14.1	16.9	19.7	22.5	25.3
Power (dBFS)	-4.1	-4.6	-6.2	-9.1	-14.3	-24.1	-22.0	-17.4	-16.6	-18.1

(a) Using linear interpolation between the two samples that bracket the half-power level, find the half-power beamwidth. The trace is symmetric about $0°$.

$\theta_{\text{HP}} =$ $12.9°$

Half power is 3 dB below the peak, so the level is $-4.1 - 3.0 = -7.1$ dBFS. That falls between the samples at $5.6°$ (-6.2) and $8.4°$ (-9.1).

$$\theta_{-3} = 5.6 + 2.8\left(\frac{7.1 - 6.2}{9.1 - 6.2}\right) = 5.6 + 2.8(0.31) = 6.5°$$

The trace is symmetric, so the beamwidth is twice this.

$$\theta_{\text{HP}} = 2(6.5) = 12.9°$$

Theory gives $13.2°$, so the sweep reads about three tenths of a degree narrow.

(b) Find the first-null beamwidth as the sweep reports it, and compare it with FNBW $= 2\arcsin(\lambda/Nd)$.

$\text{FNBW}_{\text{meas}} =$ $28.2°$

$\text{FNBW}_{\text{calc}} =$ $30.1°$

The deepest sample between the main lobe and the sidelobe is -24.1 dBFS at $14.1°$, so the sweep reports

$$\text{FNBW}_{\text{meas}} = 2(14.1) = 28.2°$$

With $\lambda = 29.1$ mm and $Nd = 8(14) = 112$ mm,

$$\text{FNBW} = 2\arcsin\left(\frac{29.1}{112}\right) = 2\arcsin(0.2598)$$
$$= 2(15.1°) = 30.1°$$

The measured value is $1.9°$ narrow because the true null at $15.1°$ falls between the samples at $14.1°$ and $16.9°$. The sweep can only report a null at an angle it actually visited.

(c) Find the first sidelobe level relative to the peak, in dBc.

SLL = -12.5 dBc

The tallest sample beyond the null is -16.6 dBFS at $22.5°$. Relative to the peak,

$$\text{SLL} = -16.6 - (-4.1) = -12.5 \text{ dBc}$$

which is within a few tenths of the -12.8 dB the array factor gives for eight uniform elements.

(d) No sample in the sweep reads below about -24 dBFS, which is 20 dB below the peak, yet the noise

floor on this kit is 23 dB below the peak. Explain in one or two sentences why the deepest sample sits 3 dB above the floor rather than on it.

The receiver reports signal power plus noise power. At the sample nearest the null the array factor has fallen to about -23 dB, which is the floor level itself, so the two contributions are equal and their sum is 3 dB above either one. A measured null depth is therefore a statement about the dynamic range of the measurement rather than about the array.

2. **Predict before you measure.** The same array is reconfigured by setting some element gains to zero. The spacing $d = 14$ mm and the frequency 10.3 GHz do not change, so $\lambda = 29.1$ mm throughout.

   (a) Rx1, Rx2, Rx7 and Rx8 are set to zero, leaving Rx3 through Rx6 active — four adjacent elements. Predict the half-power beamwidth from $\theta_{\text{HP}} \approx 0.886\lambda/(Nd)$.

   θ_{HP} = ≈ 26-27°

   Four active elements span $Nd = 4(14) = 56$ mm.

   $$\theta_{\text{HP}} \approx \frac{0.886(29.1)}{56} = 0.4604 \text{ rad} = 26.4^\circ$$

   Solving the array factor exactly gives 27.4°; the small-angle step inside the 0.886 rule costs about a degree at this beamwidth.

   (b) Now only Rx3, Rx4 and Rx5 are left on — three adjacent elements. Predict the half-power beamwidth and the first-null beamwidth.

   θ_{HP} = ≈ 35°

   FNBW = 87.7°

   Three adjacent elements span $Nd = 3(14) = 42$ mm.

   $$\theta_{\text{HP}} \approx \frac{0.886(29.1)}{42} = 0.6138 \text{ rad} = 35.2^\circ$$

   $$\text{FNBW} = 2\arcsin\left(\frac{29.1}{42}\right) = 2\arcsin(0.6929)$$
   $$= 2(43.8^\circ) = 87.7^\circ$$

   The beam is wide enough that the 0.886 rule is starting to under-predict; expect the sweep to read closer to 37°.

   (c) By how many dB should the peak of the three-element sweep sit below the peak of the full eight-element sweep, if every element gain that is still on remains at 100%?

   Δ = −8.5 dB

   At boresight the element signals add in voltage, so the peak field is proportional to the number of active elements.

   $$\Delta = 20\log_{10}\left(\frac{3}{8}\right) = 20\log_{10}(0.375) = -8.5 \text{ dB}$$

   (d) A student proposes reaching the same 35° beamwidth by leaving all eight elements on and turning every gain down to 37.5%. Explain in one or two sentences why that does not work.

   Beamwidth is set by the active aperture Nd, not by how much gain the elements are given. Turning all eight gains down uniformly keeps the aperture at 112 mm, so the beam stays at about 13° and only the peak level drops. To widen the beam the aperture itself has to shrink.

3. **Reconciling a sidelobe.** A second group measures the first sidelobe of the full eight-element array at -11.5 dBc, while the array factor for $N = 8$ predicts -12.8 dB. Their noise floor sits at -23 dBc and the sweep step is $2.8125°$.

   (a) The floor adds noise power to the sidelobe power the receiver reports. Treating the two as uncorrelated, compute the level the sweep should report for a true -12.8 dB sidelobe sitting on a -23 dBc floor.

   SLL = -12.4 dBc

   Convert both to power ratios and add.

   $$p_{\text{sig}} = 10^{-12.8/10} = 0.0525, \qquad p_{\text{n}} = 10^{-23/10} = 0.0050$$

   $$p_{\text{tot}} = 0.0525 + 0.0050 = 0.0575$$

   $$\text{SLL} = 10\log_{10}(0.0575) = -12.4 \text{ dBc}$$

   The floor alone accounts for about 0.4 dB of lift.

   (b) Their peak reading is also affected. The peak is 23 dB above the floor by definition. By how much does the floor lift the peak reading, and which way does that push the reported sidelobe level in dBc?

   peak lift = $+0.02$ dB

   Normalize the peak signal power to 1.

   $$p_{\text{tot}} = 1 + 0.0050 = 1.0050$$
   $$10\log_{10}(1.0050) = +0.02 \text{ dB}$$

   The peak lifts by only about 0.02 dB, which is negligible. Because the sidelobe lifts by 0.4 dB and the peak barely moves, the ratio of the two moves *up*, making the reported sidelobe look higher than the array factor predicts.

   (c) The floor explains 0.4 dB of the 1.3 dB discrepancy. Name two other contributions that plausibly make up the rest, and say for each whether it would move the reading up or down.

   First, run-to-run noise on the sidelobe sample itself. The sidelobe is only about 10 dB above the floor, so a single sweep reading scatters a few tenths of a decibel in either direction. Second, residual amplitude and phase errors across the eight elements after calibration. These break the uniform distribution slightly and raise the sidelobes, so that contribution moves the reading up. Grid straddle is a third contribution, but it moves the reading *down* and by very little here, since the sidelobe peak at $21.9°$ is caught by the sample at $22.5°$ to within 0.05 dB.

4. **The two-element case.** With only Rx4 and Rx5 active the array has two elements at $d = 14$ mm, and at 10.3 GHz the array factor is $AF = \cos(\psi/2)$ with $\psi = kd\sin(\theta)$.

   (a) Find the half-power beamwidth exactly, without using the $0.886\lambda/(Nd)$ rule.

   θ_{HP} = 62.6°

   Set the array factor to $1/\sqrt{2}$.

   $$\cos(\psi/2) = \frac{1}{\sqrt{2}} \quad \Rightarrow \quad \frac{\psi}{2} = \frac{\pi}{4} \quad \Rightarrow \quad \psi = \frac{\pi}{2}$$

   With $\psi = kd\sin(\theta)$ and $kd = 2\pi d/\lambda$,

   $$\sin(\theta) = \frac{\pi/2}{2\pi d/\lambda} = \frac{\lambda}{4d} = \frac{29.1}{4(14)} = 0.5196$$

   $$\theta = 31.3^\circ \quad \Rightarrow \quad \theta_{HP} = 62.6^\circ$$

   (b) Show that the $0.886\lambda/(Nd)$ rule fails here, and state the percentage error it makes.

   error = -15.7%

   $$\theta_{HP} \approx \frac{0.886(29.1)}{28} = 0.9208 \text{ rad} = 52.8^\circ$$

   $$\text{error} = \frac{52.8 - 62.6}{62.6} = -15.7\%$$

   The rule assumes the half-power point is at a small enough angle that $\sin(\theta) \approx \theta$. At 31° that is no longer true.

   (c) Evaluate $\lambda/2d$ and use it to explain why the lab table lists the first-null beamwidth of this array as 180°.

   $\lambda/2d$ = 1.039

   The first null of $\cos(\psi/2)$ needs $\psi = \pi$, which requires

   $$\sin(\theta) = \frac{\lambda}{2d} = \frac{29.1}{2(14)} = 1.039$$

   No real angle has a sine greater than 1, so the null lies outside visible space and no null appears anywhere in the $\pm 90^\circ$ scan. With no null to measure, the beamwidth between first nulls is quoted as the full 180° scan range by convention.

(d) How far below its own peak does this two-element pattern fall at $\theta = 90°$? Compare that with the 23 dB of dynamic range the sweep has above the noise floor at eight elements, and say what the trace will look like near the edges of the scan.

drop at $90°$ = -24.5 dB

At $\theta = 90°$, $\sin(\theta) = 1$ and $\psi = 2\pi(14)/29.1 = 3.023$ rad, so

$$|AF| = \cos(1.511) = 0.0597$$
$$20\log_{10}(0.0597) = -24.5 \text{ dB}$$

The two-element peak is itself 12 dB below the eight-element peak, so it clears the floor by only about 11 dB. The pattern needs 24.5 dB of range to reach endfire and the measurement has 11 dB, so the trace flattens into the noise well before $\pm 90°$ and the outer part of the roll-off is not visible.

5. **Design judgment.** Your section is asked to demonstrate a 20° beam on the PHASER at 10.3 GHz, using only the Element Gains controls.

   (a) How many adjacent elements must be active to give a half-power beamwidth closest to 20°? Give the aperture and the resulting beamwidth.

   N = 5 elements

   θ_{HP} = 21.1°

   Solve the beamwidth rule for the aperture.

   $$Nd = \frac{0.886\lambda}{\theta_{HP}} = \frac{0.886(29.1)}{20^\circ(\pi/180)} = \frac{25.78}{0.3491} = 73.8 \text{ mm}$$

   $$N = \frac{73.8}{14} = 5.3$$

   Element counts are integers, so take $N = 5$: $Nd = 70$ mm and

   $$\theta_{HP} \approx \frac{0.886(29.1)}{70} = 0.3683 \text{ rad} = 21.1^\circ$$

   Six elements would give 17.6°, which is further from the target.

   (b) What does this cost in peak received level compared with the full eight-element array, and how much dynamic range is left above the noise floor?

   Δ = -4.1 dB

   range = 18.9 dB

   $$\Delta = 20\log_{10}\left(\frac{5}{8}\right) = 20\log_{10}(0.625) = -4.1 \text{ dB}$$

   The floor does not move, so the range above it drops from about 23 dB to

   $$23 - 4.1 = 18.9 \text{ dB}$$

   The first sidelobe of a five-element uniform array is near -12 dB, which still clears the floor comfortably.

   (c) A classmate suggests instead applying a Hann taper to all eight elements, since that also widens the beam to about 19.5°. State one advantage and one disadvantage of the taper against switching elements off, and say which you would choose for a demonstration whose purpose is to show that beamwidth depends on aperture.

   The taper keeps the full 112 mm aperture and pushes the sidelobes below the noise floor, so the trace is cleaner and the main lobe is the only thing on the plot. Its disadvantage for this purpose is that it widens the beam by changing the shape of the amplitude distribution, not by changing the aperture, so it demonstrates the opposite of the intended point. It also costs about 4.7 dB of peak, which is no better than the five-element case. For a demonstration about aperture, switch elements off: the beamwidth and the aperture then move together and the claim is visible in one control.

Part B – Learning Objective 3.6: I can distinguish between array factor and true antenna pattern and account for element pattern effects.

6. **Reading dBc off a Tracking-tab trace.** A cadet walks an HB100 around the PHASER on a 1 m arc with the beam held at broadside and records the Signal vs Time trace. The three lobe peaks read -18.4 dBFS (main lobe), -31.2 dBFS (left first sidelobe), and -29.9 dBFS (right first sidelobe).

   (a) Convert both sidelobe readings to dBc.

   A dBc value is the lobe peak referred to the main-lobe peak, so subtract:

   $$L_{\text{left}} = -31.2 - (-18.4) = -12.8 \text{ dBc}$$
   $$L_{\text{right}} = -29.9 - (-18.4) = -11.5 \text{ dBc}$$

   L_{left} = -12.8 dBc

   L_{right} = -11.5 dBc

   (b) The cadet's partner raises **Rx Gain** by 6 dB and repeats the walk. State what happens to the three dBFS readings and what happens to the two dBc values, and explain why the two answers differ.

   All three dBFS readings rise by about 6 dB as well, because the gain change multiplies every received sample by the same factor. The dBc values do not change, because each one is a difference between two readings taken through the same gain setting and the common term cancels in the subtraction. This is why a pattern is reported as a ratio to its own peak: the result does not depend on the receiver's absolute scale, which is not calibrated here anyway.

   (c) A uniform 8-element array has its first sidelobes at -13 dBc. Comment on the pair measured above: is this array uniformly illuminated, and what accounts for the 1.3 dB spread between the two sides?

   Both readings straddle -13 dBc within about a decibel and a half, which is consistent with uniform illumination measured in a room. A tapered illumination would put the sidelobes at least 17 dB down, and a badly calibrated array would show a much larger imbalance. The 1.3 dB spread between left and right comes from the measurement rather than the array: classroom multipath adds roughly ± 1 dB of ripple, and the walking speed and radius were not identical on the two halves of the arc. The array factor of a symmetric uniform array is symmetric at broadside, so the asymmetry has no home in the physics of the array itself.

7. **Why the time axis is not an angle axis.** The same trace runs from $t = 0.00$ s, when the source was at $-90°$, to $t = 7.40$ s, when it reached $+90°$. The main-lobe peak lands at $t = 3.10$ s. The beam was commanded to broadside.

   (a) Explain why the lobe *amplitudes* on this trace are usable while the lobe *angles* are not.

   The amplitude recorded at any instant is the pattern evaluated at whatever angle the source occupied at that instant, and that value does not depend on how fast the source was moving. Walking slowly through a lobe spends more samples on it and draws it wider on the plot, but the height each sample reaches is unchanged. The angle, in contrast, is never recorded at all. The x-axis stores elapsed time, and recovering angle from time requires knowing the walking speed at every instant, which nothing in the measurement captures.

   (b) Assume a constant walking speed and convert the main-lobe time to an angle. Compare it against the commanded beam angle and state the error.

   A constant speed maps time linearly onto the $180°$ swept:

$$\theta = -90° + 180° \left(\frac{3.10}{7.40}\right)$$
$$= -90° + 75.4° = -14.6°$$

   The beam was commanded to $0°$, so the linear map is in error by $14.6°$.

   θ = $-14.6°$

   error = $14.6°$

   (c) The true half-power beamwidth is $13.1°$. If the hand moves as much as 30% faster or slower than its average while crossing the main lobe, give the range of beamwidths a constant-speed conversion would report.

   A hand moving 30% fast spends $1/1.30$ of the expected time in the lobe, and a constant-speed map converts that shortened interval into a narrower angle. A hand moving 30% slow does the opposite:

$$\theta_{\text{fast}} = \frac{13.1°}{1.30} = 10.1°$$
$$\theta_{\text{slow}} = \frac{13.1°}{0.70} = 18.7°$$

   The reported beamwidth spans nearly a factor of two on a pattern that never changed.

   range = $10.1°$ to $18.7°$

   (d) Name the piece of range hardware that removes this error and state what it records that the hand walk does not.

   A turntable with a shaft encoder. The turntable carries the antenna through a commanded rotation, and the encoder tags every recorded sample with the angular position it was taken at, typically to a small fraction of a degree. The measurement then stores angle directly instead of inferring it from time, so no assumption about rotation speed is needed and a non-uniform rotation costs nothing.

8. **Scan loss and sidelobe asymmetry.** The projected-aperture rule puts the peak power gain of a steered array at $\cos\theta_0$ relative to broadside.

   (a) Compute the predicted peak drop at steer angles of $30°$, $45°$, and $60°$.

   The rule applies to power, so the drop in dB is $10\log_{10}(\cos\theta_0)$:

   $$30° : \ 10\log_{10}(0.866) = -0.6 \text{ dB}$$
   $$45° : \ 10\log_{10}(0.707) = -1.5 \text{ dB}$$
   $$60° : \ 10\log_{10}(0.500) = -3.0 \text{ dB}$$

   $30°$ = -0.6 dB

   $45°$ = -1.5 dB

   $60°$ = -3.0 dB

   (b) A broadside walk peaks at -18.4 dBFS and a $30°$ walk peaks at -19.3 dBFS. Give the measured scan loss, and state whether one pair of walks resolves it given ± 0.5 dB of run-to-run scatter.

   $$\Delta = -19.3 - (-18.4) = -0.9 \text{ dB}$$

   Δ = -0.9 dB

   The measured 0.9 dB sits 0.3 dB from the predicted value of 0.6 dB and well inside the ± 0.5 dB scatter of a hand walk. One pair of runs is therefore consistent with the prediction but does not resolve it: the same pair of numbers would also be consistent with a scan loss anywhere between about 0.4 and 1.4 dB of scan loss. Averaging several walks at each steer angle narrows the scatter enough to make the comparison worth reporting.

   (c) The measured $30°$ trace shows first sidelobes of -12.2 dBc on the low-angle side and -15.2 dBc on the high-angle side. Explain why they are unequal, given that the array factor alone would make them equal.

   The array factor of a uniformly excited array is a function of $\sin\theta - \sin\theta_0$, so its sidelobes are equal in that variable and the two first sidelobes of a steered beam carry the same array-factor height. The radiated pattern is the array factor multiplied by the element factor, and the element factor falls off monotonically away from broadside. The high-angle sidelobe near $+59°$ therefore sits where the element pattern is much weaker than at the low-angle sidelobe near $+8°$, and the element factor pulls it down by the difference. Asymmetry in a steered pattern is evidence of the element factor, not of a fault in the array.

(d) Real patch elements follow a power pattern nearer $\cos^{1.4}\theta$ than $\cos\theta$. Recompute the 60° scan loss and state how much worse it is than the ideal-element bound.

loss = -4.2 dB

$$10\log_{10}\left(\cos^{1.4} 60^\circ\right) = 14\log_{10}\left(0.500\right)$$
$$= -4.2 \text{ dB}$$

That is 1.2 dB worse than the -3.0 dB ideal-element bound. The $\cos\theta$ rule is a best case set by the projected aperture alone; a real element's own pattern adds to it, and a link budget that scans to 60° should carry the larger number.

9. **How far out is far enough.** The PHASER's 8 elements are spaced 14 mm apart, so the aperture spans $D = 98$ mm between the outer element centers. The HB100 runs at 10.525 GHz.

   (a) Find the far-field distance for this aperture and state whether the 1 m arc used in lab is far enough.

   $r =$ 0.67 m

   $$\lambda = \frac{3 \times 10^8}{10.525 \times 10^9} = 0.0285 \text{ m}$$
   $$r \geqslant \frac{2D^2}{\lambda} = \frac{2(0.098)^2}{0.0285} = 0.67 \text{ m}$$

   The 1 m arc is comfortably outside the 0.67 m boundary, so the pattern has settled into its far-field shape everywhere along the walk.

   (b) A Ku-band panel with a 0.200 m aperture is to be measured at 15.0 GHz. Find its far-field distance.

   $r =$ 4.00 m

   $$\lambda = \frac{3 \times 10^8}{15.0 \times 10^9} = 0.0200 \text{ m}$$
   $$r \geqslant \frac{2(0.200)^2}{0.0200} = 4.00 \text{ m}$$

   Doubling the aperture and halving the wavelength pushes the requirement out by a factor of eight.

   (c) The Ku-band panel is measured on the same 1.0 m classroom arc anyway. Find the phase error across the aperture edge and compare it against the 22.5° tolerance behind the far-field criterion.

   $\Delta\phi =$ 90°

   The extra path to the aperture edge relative to its center is $\Delta \approx D^2/8r$, and the phase error is $360^\circ \Delta/\lambda$:

   $$\Delta = \frac{(0.200)^2}{8(1.0)} = 0.005 \text{ m}$$
   $$\Delta\phi = \frac{360^\circ(0.005)}{0.0200} = 90^\circ$$

   That is four times the 22.5° the criterion allows.

   (d) Describe what a 90° edge phase error does to the measured pattern, and say which pattern feature it corrupts worst.

   The edges of the aperture arrive out of phase with the center, so the measurement no longer sees a plane wave and the aperture is effectively defocused. The main lobe broadens and its peak drops, but the damage is worst in the nulls and the sidelobes: the cancellation that produces a deep null depends on precise phase alignment across the aperture, so the nulls fill in and the sidelobes read several decibels high. A sidelobe level measured this way is pessimistic, and a null depth measured this way is meaningless.

10. **Classroom against chamber.** The lab trace carries about ± 1 dB of ripple from room reflections adding to the direct path.

(a) List which of the four hand-measurement artifacts from the lab an anechoic chamber removes, and which one it does not remove by itself.

A chamber removes the reflection artifact directly: absorber on every surface leaves only the direct path, so the ripple disappears. A chamber's fixtures also remove the angle error and the amplitude instability, because the antenna sits on an encoded turntable and the source is bolted at a fixed range rather than carried. The far-field condition is not removed by the absorber at all; it is a geometric requirement on the separation distance, and a chamber satisfies it only if the chamber is long enough for the aperture and frequency being measured. A large antenna in a short chamber has the same problem as a large antenna in a classroom.

(b) A single reflected path of relative amplitude ρ adds to a direct path of amplitude 1. Find the ρ that produces a $+1$ dB peak in the received amplitude, and express it in dB.

$\rho =$ $0.122 = -18.3$ dB

The two paths add in phase at the ripple peak:

$$20\log_{10}(1+\rho) = 1.0 \text{ dB}$$
$$1+\rho = 10^{0.05} = 1.122$$
$$\rho = 0.122 \rightarrow -18.3 \text{ dB}$$

A reflection only 18 dB below the direct path is enough to move the reading by a decibel, which is why room reflections matter at levels that sound negligible.

(c) The main-lobe peak and the sidelobe peak each carry their own ± 1 dB of ripple. Give the worst-case uncertainty on a sidelobe level reported in dBc, and give the range a true -13 dBc sidelobe could read.

uncertainty = ± 2 dB

range = -11 to -15 dBc

A dBc value is a difference of two readings, and in the worst case the two errors have opposite signs and add:

$$\pm 1 \text{ dB} + \pm 1 \text{ dB} = \pm 2 \text{ dB}$$

A true -13 dBc sidelobe can therefore be reported anywhere from -11 to -15 dBc without anything being wrong with the array.

(d) A customer requires written proof that this array's sidelobes are below -25 dBc. State whether the classroom measurement can support that claim, and give the two reasons that decide it.

It cannot. The first reason is the uncertainty computed above: a ± 2 dB window on every dBc reading cannot certify a level that must sit below a specific number, because a measured -26 dBc could be a true -24 dBc. The second reason is dynamic range. The electrical sweep's noise floor sits about 23 dB below the peak, so a -25 dBc sidelobe is buried in the floor and there is nothing to read. Both problems are what a chamber and a calibrated receiver are for, and the claim needs a range measurement.

Documentation: _____